Background-contrastive neighbor embedding: method, evidence and open questions

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1 Summary

This report describes the current state of the method, the reasoning and evidence behind each design decision, what it has been tested on, and what remains open.

The method. BC-NE embeds a target dataset X_T in two dimensions while suppressing variation that it shares with a background dataset X_B. Its core is a *background-suppressed neighbor graph*. For each target point, the method takes its nearest background points, subtracts their mean and their top few principal directions, and builds the k-nearest-neighbor (kNN) graph on what remains. The graph is then embedded with two-stage Neg-t-SNE, a negative-sampling neighbor embedding [3]. Optionally, the method adds *background negatives*: extra repulsion between target points whose background neighborhoods overlap. We write BC-NE for the method without background negatives and BC-NE+BN for the method with them.

Key findings.

1. **The residual graph is what makes the method work.** MNIST-on-grass is a real-image benchmark: MNIST digits overlaid on grass photographs. On it, BC-NE+BN beats the method's earlier form, which used background negatives on the raw kNN graph, on every confirmation seed. Digit kNN accuracy is 0.973 vs 0.931 *(pre-registered)*. With all ten digits the gap is wider: 0.530 vs 0.226 *(pre-registered)*. Without background negatives, the residual graph alone raises plain Neg-t-SNE by 0.10–0.13 *(descriptive)*.
2. **Background negatives add little once the graph is background-suppressed.** With two digits, BC-NE+BN and BC-NE differ by at most 0.002 per seed, with one exact tie *(pre-registered)*. With ten digits the difference is small (+0.016 on average), and we do not treat it as a real gain *(interpretation)*.
3. **On MNIST-on-grass with two digits the method ties contrastive PCA (cPCA) followed by t-SNE, and is more stable across seeds.** Neither is consistently better *(pre-registered)*. Across seeds, BC-NE+BN ranges over 0.972–0.974 and cPCA→t-SNE over 0.941–0.979 *(descriptive)*.
4. **Global cPCA wins when the background is additive.** If target = signal + independent background, the difference of the target and background covariances is exactly the signal covariance *(our derivation)*. Accordingly, cPCA-based pipelines beat raw-graph background negatives on every additive benchmark we ran: synthetic *(descriptive)* and real *(pre-registered)*. With ten digits, cPCA→t-SNE at its label-free α falls behind the method (0.433 vs 0.530) *(pre-registered)*, but with α tuned on labels it is ahead (0.689) *(descriptive)*. Tuning α that way needs labels, which real data does not have, so it is not a realistic setting *(interpretation)*.

Main open question. The residual graph has so far been tested only on MNIST-on-grass. That dataset's background is additive, so global cPCA already does about as well. The residual graph's intended advantage is suppressing a background that no single global projection can remove. That advantage has not yet been tested on real data with a non-additive background (section 7).

How to read this report. Section 2 sets out the problem, data and evaluation; sections 3 and 4 give the background and the method; section 5 covers the design decisions and their evidence; section 6 presents the full results; and sections 7 and 8 cover what is untested and what comes next.

Every claim carries one of five labels:

- *(published)*: a cited result;
- *(our derivation)*: derived here, not taken from the literature;
- *(pre-registered)*: a verdict under a rule committed before the results existed;
- *(descriptive)*: a measurement with no verdict attached;
- *(interpretation)*: our explanation, not directly tested.

2 Problem, data and evaluation

2.1 Problem and notation

We have a *target* dataset $X_T \in \mathbb{R}^{n_T \times D}$ and a *background* dataset $X_B \in \mathbb{R}^{n_B \times D}$ in the same feature space. The background contains only nuisance variation. The target contains the same kind of nuisance variation plus a target-specific *signal*. The goal is a two-dimensional embedding of the target points whose neighborhoods reflect the signal rather than the background. This is the neighbor-embedding analogue of contrastive PCA [1]. cPCA finds directions with high target variance and low background variance: the top eigenvectors of $C_T - \alpha C_B$ for a contrast parameter $\alpha \geq 0$.

Notation.

- **Embedding:** $y_i \in \mathbb{R}^2$ is the embedding of target point i, and $d_{ij} = \|y_i - y_j\|$. The Cauchy kernel is $\phi(d) = 1/(1+d^2)$, and $q_{ij} = \phi(d_{ij})$ is the model similarity.
- **Graph and sampling:** p_{ij} is the uniform distribution over the directed edges of the symmetrized kNN graph of the targets ($k = 15$). ξ is a noise distribution over pairs. m and m_B are the numbers of uniform and background negatives per edge.
- **Residual graph:** k_{loc} is the number of background neighbors used to form the residual, and r the number of local background directions removed.
- **Covariances:** C_T and C_B are the covariances of X_T and X_B, and C_S and C_G are those of the signal and background components. Q^T is the transpose of Q.

2.2 Datasets

Synthetic benchmarks. The synthetic benchmarks share one template: $n_T = n_B = 5000$ points in $D = 50$ dimensions, with isotropic noise of standard deviation 0.1.

- **Background:** six equal-weight Gaussian clusters, with centers on a regular simplex of edge 30 in a 5-dimensional latent space.
- **Signal:** four groups, with centers on a simplex of edge 6 in a 3-dimensional latent space, scaled by a strength ρ.
- **Choice of ρ:** the largest value at which plain Neg-t-SNE still hides the signal. A benchmark is *valid* only if plain Neg-t-SNE leaves signal kNN accuracy at or below 0.50 and background accuracy at or above 0.90, while the clean signal alone is separable (accuracy at least 0.95).

Table 1 lists the benchmarks. Labels for signal group and background cluster are used only for evaluation.

MNIST-on-grass. This follows the construction of the cPCA paper [1]:

- **Images:** MNIST digits 0 and 1 [7], at 25% intensity, added to random 28×28 grayscale patches cut from 1274 ImageNet grass photographs [4].
- **Target and background:** drawn from two disjoint halves of the photographs.
- **Seeds:** each seed has 5000 target and 5000 background images.
- **Recipe:** the authors' notebook, with five documented changes made for determinism and provenance.

Table 1: The benchmarks used in this report.

name	construction	what it tests
additive (orthogonal, overlapping, nonlinear)	$x_T = \rho A z_s + B(z_b) + \varepsilon$, $x_B = B(z_b') + \varepsilon$. Orthogonal: signal and background spans orthogonal. Overlapping: half the signal span inside the background span. Nonlinear: B a random tanh network. $\rho = 0.2, 0.2, 0.3$.	whether background negatives beat simple controls
MNIST-on-grass (real)	digits 0/1 (and 0–9) at 25% intensity added to 28×28 grass patches; background = grass patches	real images, additive background

- **Ten digits:** a second version uses all ten digits, 500 of each per seed (section 6.3).

The grass is added independently of the digit, so this dataset is additive.

2.3 Evaluation protocol

Pre-registration and seeds. Every experiment was pre-registered: its runs, decision rules and predictions were committed to the repository before the results they govern (section B). Each dataset is generated from a seed, which also seeds every run on it. Settings such as m_B are chosen on *selection seeds* 0–2. Verdicts use fresh *confirmation seeds*, usually 3–5.

The verdict rule. Runs are paired by seed. "X beats Y, consistent across 3 paired seeds" means that the mean difference is positive *and* every per-seed difference is positive. Anything else is reported as "not consistent", and no p-values are reported. In every experiment, reruns were checked to be bit-identical, and runs were flagged for collapse or fragmentation (none was flagged).

Metrics. Both metrics are computed on the 2-D embedding.

- *Signal kNN accuracy*: a 15-nearest-neighbor classifier predicting the signal label, with stratified 5-fold cross-validation. On MNIST-on-grass this is *digit kNN accuracy*.
- *Recall vs grass*: the fraction of a point's 15 embedding neighbors that are also among its 15 nearest images by grass patch. Lower is better.

Controls.

control	rules out that the gain comes from
plain Neg-t-SNE	(baseline)
uniform extra negatives (same number, drawn uniformly)	more repulsion
t-SNE-end spectrum reference ($\bar{Z} = 100n$, i.e. $c = 500/n$)	a shift along the attraction–repulsion spectrum
cPCA→Neg-t-SNE, cPCA→t-SNE	linear background suppression
t-SNE, PCA→t-SNE	standard practice

3 Background: neighbor embeddings as contrastive learning

t-SNE [12] and UMAP [9] can both be written as attraction along kNN edges plus repulsion between other pairs. In t-SNE the repulsion comes from normalizing q over all pairs; in UMAP it comes from sampled noise pairs *(published)* [3]. The balance between the two forces places a method on an attraction–repulsion spectrum, running from t-SNE-like to UMAP-like embeddings *(published)* [2].

Damrich et al. [3] make this precise through contrastive learning. Both estimators below sample positive pairs from p and noise pairs from ξ, and train a classifier to tell them apart.

- **Noise-contrastive estimation (NCE).** The classifier is $q_\theta/(q_\theta + m\xi)$. At the optimum $q_\theta = p$, whatever the full-support noise distribution *(published)* ([5]; Theorem 1 of [3]).

- **Negative sampling** [10, 8]. The classifier is $q_\theta/(q_\theta+1)$: the noise probability does not appear in it. Corollary 2 of [3] shows that, with uniform noise over the $|X|$ pairs, negative sampling learns a rescaled distribution $q = (|X|/m)\,p$ *(published)*.

Neg-t-SNE is negative sampling with the Cauchy kernel. In the `contrastive-ne` package its classifier is $q/(q+c)$ for a constant c, with default $c=1$. Moving c moves the embedding along the spectrum *(published)* [3].

Extension to a general noise distribution *(our derivation)*. Corollary 2 assumes uniform noise. Background negatives make the noise non-uniform, so we need the optimum for a general ξ. The expected negative-sampling loss with classifier $q/(q+c)$ is

$$\mathcal{L}(q) = \sum_x \Big[-p(x)\log\frac{q(x)}{q(x)+c} \;-\; m\,\xi(x)\log\frac{c}{q(x)+c}\Big], \tag{1}$$

where the sum runs over pairs $x=(i,j)$. Treat each $q(x)$ as a free variable and set the derivative to zero:

$$-\frac{p}{q} + \frac{p+m\xi}{q+c} = 0 \quad\Longrightarrow\quad q^*(x) = \frac{c\,p(x)}{m\,\xi(x)}. \tag{2}$$

For $p(x)>0$ the loss tends to $+\infty$ both as $q\to 0$ and as $q\to\infty$, so this stationary point is the minimum. With uniform $\xi = 1/|X|$ and $c=1$, eq. (2) recovers Corollary 2.

With background negatives, each edge with head i draws m uniform negatives and m_B negatives from $\xi_B(i,\cdot)$, the latter weighted by γ in the loss. The expected noise weight on a pair is then $m\,\xi_U(i,j) + \gamma\,m_B\,\tilde{\xi}_B(i,j)$, where $\tilde{\xi}_B(i,j) = p_i\,\xi_B(i,j)$ and p_i is the head marginal of p. The same calculation gives

$$q^*_{ij} = \frac{c\,p_{ij}}{m\,\xi_U(i,j) + \gamma\,m_B\,\tilde{\xi}_B(i,j)}. \tag{3}$$

Pairs that the background explains, i.e. those with large $\tilde{\xi}_B$, receive a smaller target similarity, so they are pushed apart. This includes kNN edges whose endpoints are close for background reasons.

The derivation is idealized. It assumes:

- that each q_{ij} can be set freely, which a 2-D embedding cannot do exactly;
- that expectations replace sampling;
- that the package's in-batch uniform negatives are uniform over pairs;
- no clamping (the package clamps the classifier output to $[10^{-4}, 1]$).

UMAP. Lemma 3 of [3] shows that UMAP is negative sampling with the model $1/d_{ij}^2$ in place of the Cauchy kernel. Its repulsion therefore diverges as $d\to 0$, while Neg-t-SNE's stays bounded *(published)*.

4 The method

Algorithm 1 summarizes the full pipeline. This section specifies each step.

4.1 Background-suppressed (residual) neighbor graph

For each target point x_i we find its k_{loc} nearest background points $N_B(i) \subset X_B$ by exact Euclidean search. Let μ_i be their mean, and let $V_i \in \mathbb{R}^{D\times r}$ be the top r right singular vectors of the centered neighbors $\{b-\mu_i : b\in N_B(i)\}$. The residual of x_i is

$$\tilde{x}_i = (I - V_i V_i^T)(x_i - \mu_i). \tag{4}$$

The attraction graph is the exact $k=15$ nearest-neighbor graph of the residuals, symmetrized by union and with binary weights.

The defaults are $k_{\text{loc}}=30$ and $r=2$. They were selected on one dataset, and the choice barely mattered there (section 5.6). The residual is invariant to sign flips of the columns of V_i. It needs neither a clustering of the background nor a global projection. Only the graph changes: the initialization and the background negatives, if used, are computed on the raw features.

4.2 Two-stage Neg-t-SNE

We use Neg-t-SNE [3] as implemented in `contrastive-ne` v0.4.0, non-parametric, with the package defaults:

- $m = 5$ negatives per edge, drawn from the current minibatch;
- $c = 1$;
- minibatches of 2^{15} directed edges;
- SGD with learning rate 1, annealed linearly to 0;
- elementwise gradient clipping at ± 4.

Every embedding is initialized from the first two principal components, scaled so that the first has standard deviation 1. This initialization is recommended for preserving global structure *(published)* [6, 3].

Training is two separate fits on the same graph. Stage 1 runs 250 epochs from the PCA initialization. Stage 2 runs 500 epochs starting exactly from the Stage 1 embedding, with a fresh learning-rate annealing. Background negatives, when used, are active in Stage 2 only.

We use two separate fits rather than the package's built-in early-exaggeration phase, which does nothing at its negative-sampling defaults *(descriptive)*. They give an exact, reproducible handoff. The warm-up was meant to let a global layout form before the background term acts *(interpretation)*. Without background negatives it turns out to be unnecessary (section 5.7); with them, no single-stage schedule has been compared.

4.3 Optional background negatives

The background-negative distribution ξ_B pairs target points that look similar for background reasons. It is built in four steps.

1. **Background neighbors.** For each target point i, find its $k_B = 15$ nearest background points $N_B(i)$.
2. **Scores.** Score each pair of target points by the Jaccard overlap $s_B(i, j) = |N_B(i) \cap N_B(j)|/|N_B(i) \cup N_B(j)|$.
3. **Partners.** Keep the top $L = 30$ partners of each point. Ties with the 30th are kept, up to $2L$; beyond that cap, a seeded random draw decides, never target distance.
4. **Distribution.** Set $\xi_B(i, \cdot) \propto s_B(i, \cdot)$. Target kNN edges are not excluded.

For each kNN edge with head i, Stage 2 adds m_B negatives $b \sim \xi_B(i, \cdot)$, with the loss term

$$-\gamma \sum_{b=1}^{m_B} \log\Big(1 - \frac{\phi_{ib}}{\phi_{ib} + c}\Big), \qquad \gamma = 1. \tag{5}$$

The term uses the same c, kernel and clamping as the package's own negatives. A unit test confirms that one background negative and one uniform negative at the same distance contribute identical loss and gradient.

By eq. (3), the term lowers the target similarity of pairs with overlapping background neighborhoods *(our derivation)*. Loosely, the embedding learns "target similarity divided by background similarity" *(interpretation)*.

4.4 Implementation

- **Package changes:** an 11-line hook in the vendored package lets the trainer add a term to each batch loss.
- **Graphs:** all kNN graphs are exact and built once per dataset. The package's own approximate graph is unseeded, so it is not used.
- **Determinism:** every fit runs with deterministic algorithms and seeded generators. The background sampler has its own generator, so with background negatives turned off the result is bit-identical to the unmodified package.

Computational cost *(our derivation)*. With n targets, n_B background points and D features:

- **Neighbor searches:** t-SNE, UMAP and Neg-t-SNE each need one target-to-target kNN search. BC-NE adds one target-to-background search (k_{loc} neighbors), and background negatives one more (k_B).

Algorithm 1 BC-NE (with optional background negatives)

Require: targets X_T, background X_B; $k_{\text{loc}} = 30$, $r = 2$, $k = 15$; optional m_B

for each target x_i **do**
  $N_B(i) \leftarrow$ the k_{loc} nearest rows of X_B to x_i
  $\mu_i \leftarrow$ mean of $N_B(i)$; $V_i \leftarrow$ top-r right singular vectors of $N_B(i) - \mu_i$
  $\tilde{x}_i \leftarrow (I - V_i V_i^T)(x_i - \mu_i)$
end for
$G \leftarrow$ exact k-NN graph of $\{\tilde{x}_i\}$, union-symmetrized, binary
$Y_0 \leftarrow$ first two principal components of X_T, scaled so the first has std 1
$Y_1 \leftarrow$ Neg-t-SNE(G, init Y_0, 250 epochs)
if $m_B > 0$ **then**
  build ξ_B from Jaccard overlaps of the k_B-nearest background sets (raw features)
end if
$Y \leftarrow$ Neg-t-SNE(G, init Y_1, 500 epochs, plus m_B negatives per edge from ξ_B if $m_B > 0$)
return Y

With exact brute-force search, as used here, each costs $O(n(n + n_B)D)$. Approximate search, as the standard packages use, applies to them equally.

- **Residuals:** one SVD of a $k_{\text{loc}} \times D$ matrix per target, $O(n\, k_{\text{loc}}^2 D)$, linear in n.
- **Training:** each epoch costs $O(nk(m + m_B))$, linear in n, as for Neg-t-SNE and UMAP.
- ξ_B**:** overlaps are counted through shared background neighbors. This is linear in n unless some background point is a neighbor of a growing number of targets.

The method is therefore in the same complexity class as Neg-t-SNE, UMAP and t-SNE: the neighbor searches dominate, and everything else is linear in n. In practice ($n = 5000$, $D = 784$, one CPU thread; table 5), BC-NE took about as long as plain Neg-t-SNE (62–79 s vs 68–75 s), so the residual graph adds little. Background negatives with $m_B = 5$ roughly doubled the time (139–151 s), and openTSNE's t-SNE was faster (34–41 s) *(descriptive)*.

5 Design decisions and their evidence

This section takes each design decision in turn: what was chosen, the alternatives, and the evidence. The experiments behind the numbers are presented in full in section 6.

5.1 Neg-t-SNE rather than UMAP or t-SNE

The method needs an embedding whose optimum responds to the noise distribution.

- **Negative sampling** has this property, because the noise distribution is absent from its classifier (eq. (2)) *(our derivation)*.
- **NCE** does not: its optimum is p for any noise *(published)*.
- **t-SNE** has no noise distribution at all.

Among negative-sampling methods we rejected UMAP, because the repulsion of its effective $1/d^2$ model diverges at short range *(published)*. Short range is exactly where background negatives act: on pairs that start close together because they share a background *(interpretation)*.

The choice has a cost. Without the background term, Neg-t-SNE scored below t-SNE on MNIST-on-grass: digit accuracy 0.858, against 0.884 for t-SNE and 0.891 for PCA→t-SNE *(descriptive)*. Comparisons with t-SNE-based controls therefore also change the embedding algorithm (section 7).

5.2 Putting the background into the noise distribution

The method's first version changed only the repulsion. It added background negatives on top of the raw kNN graph. This works only because negative sampling keeps the noise out of its classifier; under NCE, the noise distribution would cancel at the optimum *(published)*.

On the three additive synthetic benchmarks, this version beat the best uniform-extra-negatives control on every confirmation seed *(pre-registered)*. The mean gain in signal accuracy was +0.51 (orthogonal), +0.18 (overlapping) and +0.44 (nonlinear). It also beat the spectrum reference at every level *(pre-registered)* (table 2).

The uniform control is weak, however *(descriptive)*. With six equal, well-separated background clusters, about 5 in 6 uniform negatives land in another cluster. That cluster is already far away, so they add little force within a cluster. Beating this control shows that the repulsion must be targeted, not that ξ_B's targeting is the only kind that works.

5.3 Background similarity from shared background neighbors

ξ_B pairs share their background but not their signal *(descriptive)*.

- **Synthetic data:** the two points of a pair belong to the same background cluster 100% of the time. The excess of same-signal pairs over chance is at most +0.064.
- **MNIST-on-grass:** the same-digit excess is +0.055 to +0.074. The grass patches of a pair are 3.8–4.0 apart, against 7.9–8.1 for random pairs.

5.4 Moving the background into the graph

Background negatives change only the repulsion; the attraction still follows the raw kNN graph. Two measurements pointed to that graph as the limit *(descriptive)*:

- **Additive benchmarks:** none of about 925,000 kNN edges joins two background clusters, so attraction can never bring the clusters together.
- **MNIST-on-grass:** 44% of a target's raw-graph neighbors are also among its 15 nearest images by grass patch.

The "graph ceiling" is not strict, however. On MNIST-on-grass, raw-graph background negatives reached a digit accuracy of 0.931. That is above the raw graph's own digit accuracy on the same seeds, 0.901 *(descriptive)*. The main evidence for moving the background into the graph is therefore the residual-graph result itself: BC-NE+BN beats raw-graph background negatives on every MNIST-on-grass confirmation seed *(pre-registered)*.

5.5 A local residual graph rather than a global cPCA graph

The population identity behind cPCA. This is a standard argument *(our derivation)*. Suppose the target is $x_T = s + g + \varepsilon$ and the background is $x_B = g' + \varepsilon'$, where:

- s, g and ε are independent;
- g' has the same distribution as g;
- both noises have covariance $\sigma^2 I$.

Then $C_T - C_B = C_S$.

Proof. By independence the cross-covariances vanish, so $C_T = C_S + C_G + \sigma^2 I$ and $C_B = C_G + \sigma^2 I$. □

The identity holds for *any* map producing g, linear or not: only g's covariance enters, and it cancels. At $\alpha = 1$, cPCA's top directions are therefore the signal's own directions, up to finite-sample error. For $\alpha > 1$, $C_T - \alpha C_B = C_S - (\alpha - 1)(C_G + \sigma^2 I)$, so the signal survives in directions where the background has little variance *(our derivation)*.

The comparison. So on additive data the obvious alternative is a graph built on the top 10 global cPCA components. On MNIST-on-grass, that graph was the cleaner of the two *(descriptive)*:

seeds	graph	graph digit accuracy	graph grass recall
selection (0–2)	global cPCA-10	0.970	0.006
selection (0–2)	residual ($k_{\text{loc}} = 30$, $r = 2$)	0.962	0.056
confirmation (3–5)	global cPCA-10	0.957	0.006
confirmation (3–5)	residual ($k_{\text{loc}} = 30$, $r = 2$)	0.962	0.057

On the confirmation seeds the accuracy order reverses, but cPCA-10 keeps a much lower grass recall. On additive data, then, the local graph has no clear advantage. Its case rests on backgrounds that are not additive, where no single global projection removes them; that case is untested (section 7) *(interpretation)*.

5.6 Defaults $k_{\text{loc}} = 30$ and $r = 2$

Before any residual graph was computed, we committed a graph-quality gate. On each selection seed, the selected variant had to beat the better of the raw-pixel and PCA-50 graphs by at least 0.02 in graph digit accuracy. It did, by +0.025 to +0.029 *(pre-registered)*.

The choice among variants barely mattered *(descriptive)*. The grid covered $k_{\text{loc}} \in \{15, 30, 100\}$ and $r \in \{0, 1, 2, 5, 10\}$. All 15 variants reach a mean graph digit accuracy of 0.954–0.964 (0.951–0.965 on individual seeds). Every one is above both reference graphs: raw 0.904 and PCA-50 0.935. Even $r = 0$, which only subtracts the local background mean, clears both (full grid in table 7). The labels used in selection therefore decided little, which supports a fixed, label-free default *(interpretation)*. So far this holds on one dataset only.

5.7 The two-stage schedule without background negatives

Without background negatives, Stage 2 differs from Stage 1 only in the learning-rate reset. We therefore compared two-stage BC-NE with one standard 750-epoch Neg-t-SNE run (package defaults; same graph, initialization and seed) on MNIST-on-grass with two and with ten digits, seeds 0–5. Before any single run, we committed a tolerance for each dataset and measure: the larger of 0.005 and the spread over three training seeds of the two-stage run.

The single run is within tolerance on digit kNN accuracy and recall vs grass on every seed of both datasets *(descriptive)*. Over seeds 3–5, the mean change in digit accuracy is −0.0001 with two digits and −0.0016 with ten, against tolerances of 0.005 and 0.0092. On a few seeds the digit groups settle into a different arrangement, with the same quality. Without background negatives, the reset therefore does not change performance meaningfully and is unnecessary. The results in this report use the two-stage schedule. Whether the warm-up matters when background negatives act in Stage 2 has not been tested.

5.8 Current status of background negatives

With the residual graph, background negatives were redundant on MNIST-on-grass *(pre-registered)*. The smallest value on the grid, $m_B = 1$, was selected, and accuracy fell slowly as m_B grew: from 0.973 to 0.968 at $m_B = 40$ *(descriptive)* (table 8). With ten digits, BC-NE+BN was ahead of BC-NE by only +0.004 to +0.031 per seed (mean +0.016); we do not treat that as a real gain (section 6.3). They have not been tested with the residual graph on any other data.

A plausible reason for the redundancy *(interpretation)*: the residual graph and ξ_B use the same information, namely each target's nearest background points. ξ_B uses it for repulsion, the residual graph to define neighborhoods. On this dataset, putting that information into the graph is what helps.

6 Experimental results

6.1 Synthetic additive benchmarks

All synthetic results used the *raw* kNN graph with background negatives, the method's earlier form. The residual graph has not been tested on them.

Do background negatives help beyond more repulsion? Yes. Table 2 shows large gains over every control at every level, consistent across seeds *(pre-registered)*.

Does global cPCA do better? Yes, as the population identity predicts (section 5.5). cPCA→Neg-t-SNE reaches 0.980–0.989 (table 2), beating raw-graph background negatives by +0.10, +0.40 and +0.11 *(descriptive)*. It also organizes the layout by signal across clusters. Of a point's 15 embedding neighbors, a

fraction of 0.805–0.820 share its signal group but come from a different background cluster. For background negatives the fraction is 0.000–0.001 *(descriptive)*.

Table 2: Additive synthetic benchmarks: signal kNN accuracy on the confirmation seeds, mean [min, max]. Background negatives use the raw kNN graph at the m_B chosen on the selection seeds (10, 20 and 10). Every difference between background negatives and a control is consistent across 3 paired seeds *(pre-registered)*. The last column, cPCA→Neg-t-SNE, is descriptive here.

background	plain	uniform (best)	spectrum ref.	background neg.	cPCA→Neg-t-SNE
orthogonal	0.312 [0.304, 0.322]	0.387 [0.374, 0.410]	0.346 [0.337, 0.352]	0.894 [0.890, 0.899]	0.989 [0.988, 0.990]
overlapping	0.368 [0.355, 0.388]	0.405 [0.391, 0.416]	0.384 [0.382, 0.387]	0.582 [0.573, 0.588]	0.980 [0.978, 0.982]
nonlinear	0.347 [0.339, 0.353]	0.429 [0.399, 0.464]	0.381 [0.364, 0.394]	0.868 [0.843, 0.902]	0.980 [0.976, 0.984]

6.2 Real data: MNIST-on-grass

Pre-checks *(descriptive)*. The digit signal is present in the raw kNN graph: digit kNN accuracy is 0.90 on the raw images and 0.93 after PCA-50, against 1.00 on the clean digits. But the raw graph's neighborhoods also follow the grass, with grass recall 0.44.

Raw graph. Raw-graph background negatives ($m_B = 5$) beat t-SNE [11], PCA-50→t-SNE and plain Neg-t-SNE on every seed *(pre-registered)*. In turn, cPCA→t-SNE (label-free α, 10 components) beats them on every seed *(pre-registered)* (tables 3 and 4).

Residual graph. Switching to the residual graph changes the picture.

- BC-NE+BN beats raw-graph background negatives, t-SNE and PCA→t-SNE on every seed *(pre-registered)*.
- It ties cPCA→t-SNE, and it ties BC-NE without background negatives; neither comparison is consistent *(pre-registered)*.
- Recall vs grass falls from 0.022 to 0.003, the same as cPCA→t-SNE *(descriptive)*.
- In the embedding (fig. 1), the two digits form separate islands with no gradient in grass brightness. On the raw graph, by contrast, the digits split within one blob that still runs from dark to bright grass.

Checks. The BC-NE+BN seed-3 rerun was bit-identical. The raw-graph and t-SNE rows were reused from the earlier experiment, after checking that the code they depend on was unchanged and that every dataset matched its recorded checksum *(descriptive)*. The controls' embeddings are shown in fig. 3.

Table 3: MNIST-on-grass, confirmation seeds 3–5: digit kNN accuracy (mean [min, max] and per seed) and recall vs grass (lower is better). No run was flagged.

run	digit acc.	seed 3	seed 4	seed 5	recall vs grass
BC-NE+BN ($m_B = 1$)	0.973 [0.972, 0.974]	0.973	0.974	0.972	0.003
BC-NE (no background negatives)	0.972 [0.970, 0.974]	0.971	0.974	0.970	0.004
raw graph + background negatives ($m_B = 5$)	0.931 [0.926, 0.938]	0.926	0.938	0.930	0.022
plain Neg-t-SNE (raw graph)	0.858 [0.838, 0.871]	0.838	0.871	0.866	0.079
t-SNE	0.884 [0.872, 0.891]	0.872	0.891	0.890	0.135
PCA-50→t-SNE	0.891 [0.883, 0.898]	0.883	0.898	0.893	0.131
cPCA→t-SNE	0.955 [0.941, 0.979]	0.941	0.979	0.943	0.003

6.3 Real data: MNIST-on-grass with ten digits

Setup. The same recipe with all ten digits: 500 images of each digit per seed, so 5000 target and 5000 background images, with the grass pools and every setting as before. Only m_B was re-selected on seeds 0–2:

Table 4: MNIST-on-grass, pre-registered comparisons on digit kNN accuracy, confirmation seeds. Differences are the first run minus the second.

comparison	per-seed difference	mean	verdict
BC-NE+BN vs raw graph + BN	$+0.046, +0.036, +0.041$	$+0.041$	BC-NE+BN better, consistent
BC-NE+BN vs cPCA→t-SNE	$+0.032, -0.006, +0.028$	$+0.018$	not consistent
BC-NE+BN vs t-SNE	$+0.101, +0.083, +0.081$	$+0.088$	BC-NE+BN better, consistent
BC-NE+BN vs PCA→t-SNE	$+0.090, +0.076, +0.079$	$+0.082$	BC-NE+BN better, consistent
BC-NE+BN vs BC-NE	$+0.001, +0.000, +0.002$	$+0.001$	not consistent
raw graph + BN vs t-SNE	$+0.054, +0.046, +0.040$	$+0.047$	raw graph + BN better, consistent
raw graph + BN vs PCA→t-SNE	$+0.043, +0.040, +0.038$	$+0.040$	raw graph + BN better, consistent
raw graph + BN vs cPCA→t-SNE	$-0.015, -0.042, -0.013$	-0.023	cPCA→t-SNE better, consistent
raw graph + BN vs plain Neg-t-SNE	$+0.088, +0.066, +0.065$	$+0.073$	raw graph + BN better, consistent

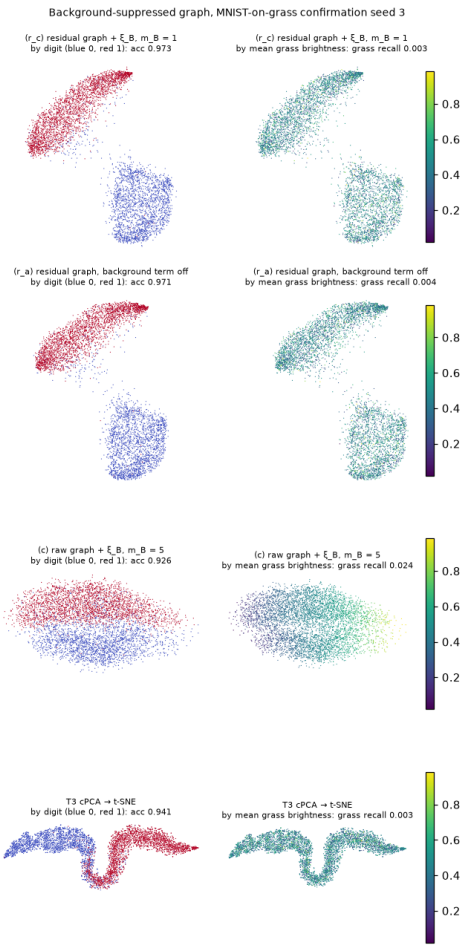


Figure 1: MNIST-on-grass, confirmation seed 3, colored by digit (left; blue 0, red 1) and by the mean brightness of each image's grass patch (right). Rows, top to bottom: BC-NE+BN (`(r_c)`); BC-NE without background negatives (`(r_a)`); raw graph + background negatives (`(c)`); cPCA→t-SNE (`T3`).

$m_B = 5$ for BC-NE+BN and $m_B = 20$ for the raw graph. The experiment was pre-registered, and chance accuracy is 0.10. As a secondary metric we add the adjusted Rand index (ARI) between the digit labels and k-means clusters of the embedding ($k = 10$, 10 restarts, seeded) *(descriptive)*.

Results (table 5 and fig. 2).

- BC-NE+BN beats t-SNE, PCA-50→t-SNE and raw-graph background negatives on every seed *(pre-registered)*. The residual graph matters even more than with two digits: 0.530 against 0.226 on the raw graph, and both raw-graph runs fall below t-SNE *(descriptive)*.
- BC-NE+BN is ahead of BC-NE on each seed, but only by +0.004 to +0.031 (mean +0.016). This passes the paired-seed rule, but with three seeds and differences this small we do not treat it as a real gain: background negatives add little *(interpretation)*.
- Against cPCA→t-SNE at its label-free α (10 components), BC-NE+BN and BC-NE both win on every seed (0.530 and 0.514 vs 0.433) *(pre-registered)*. With α and the number of components tuned on labels, cPCA→t-SNE does better than the method on these data (0.689) *(descriptive)*. Tuning α that way needs labels, which real data does not provide, so it is an optimistic bound rather than a realistic baseline; BC-NE uses no labels *(interpretation)*.
- The residual-graph and cPCA maps all remove the grass: recall vs grass is 0.003–0.004, against 0.14 for t-SNE *(descriptive)*.
- ARI ranks the runs as digit accuracy does: 0.244 for BC-NE+BN, 0.225 for BC-NE, 0.117 for cPCA→t-SNE, and at most 0.031 for every other run *(descriptive)*. It is low everywhere because no map forms ten clean clusters: in fig. 2 the digits occupy overlapping regions rather than separate islands. With two digits (computed afterwards from the saved embeddings), BC-NE+BN and BC-NE reach 0.87 and cPCA→t-SNE 0.80 *(descriptive)*.
- The absolute level is modest: 0.53 is far above chance, but the same classifier reaches 0.91 on the clean digits, so much of the digit structure is still lost *(descriptive)*.

Checks. The BC-NE+BN seed-3 and t-SNE seed-0 reruns were bit-identical, every dataset matched its checksum, and no run was flagged *(descriptive)*.

Table 5: MNIST-on-grass with ten digits, confirmation seeds 3–5: mean [min, max]. ARI is k-means ($k = 10$) against the digit labels. The 0/1 ARI is the same method on the two-digit data (table 3), each at the m_B selected there, computed afterwards from the saved embeddings. Times are wall-clock per run on one CPU thread, including graph construction. Chance accuracy is 0.10.

run	digit acc.	ARI, 0–9	ARI, 0/1 (post hoc)	recall vs grass	time per run (s)
BC-NE+BN ($m_B = 5$)	0.530 [0.521, 0.544]	0.244 [0.237, 0.248]	0.874 [0.857, 0.887]	0.003 [0.003, 0.003]	139–151
BC-NE (no background negatives)	0.514 [0.510, 0.517]	0.225 [0.205, 0.243]	0.873 [0.856, 0.887]	0.004 [0.003, 0.004]	62–79
raw graph + background negatives ($m_B = 20$)	0.226 [0.214, 0.236]	0.031 [0.026, 0.035]	0.031 [0.020, 0.040]	0.019 [0.018, 0.020]	255–276
plain Neg-t-SNE (raw graph)	0.115 [0.111, 0.119]	0.001 [0.001, 0.002]	0.005 [0.002, 0.009]	0.097 [0.095, 0.099]	68–75
t-SNE	0.253 [0.252, 0.256]	0.002 [0.001, 0.003]	0.007 [0.003, 0.009]	0.142 [0.140, 0.144]	34–41
PCA-50→t-SNE	0.255 [0.251, 0.257]	0.002 [0.002, 0.003]	0.014 [0.004, 0.030]	0.141 [0.140, 0.144]	31–33
cPCA→t-SNE (label-free α)	0.433 [0.408, 0.482]	0.117 [0.105, 0.132]	0.797 [0.733, 0.905]	0.003 [0.003, 0.004]	37–38

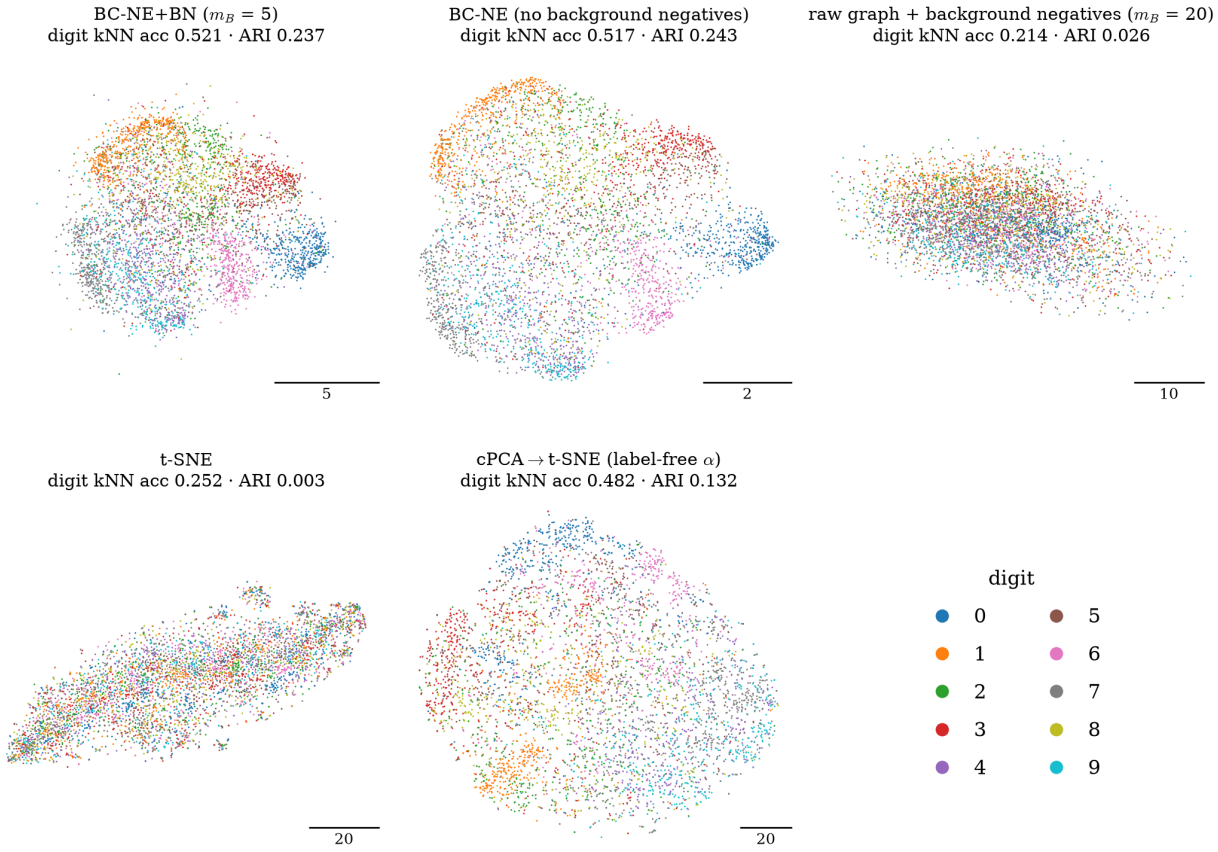


Figure 2: MNIST-on-grass with ten digits, confirmation seed 3, colored by digit. Each panel gives its digit kNN accuracy and ARI on this seed. Plain Neg-t-SNE and PCA-50→t-SNE are in table 5 only.

7 What has and has not been tested

Table 6 lists every experiment and its outcome for the method in the form tested.

Table 6: Every experiment and its outcome. "Raw" is the raw kNN graph with background negatives; "residual" is BC-NE / BC-NE+BN.

data	background	graph	competitors	outcome
3 additive synthetic	additive, discrete clusters	raw	uniform, spectrum ref., cPCA	beats controls; cPCA better
MNIST-on-grass	real, additive	raw	t-SNE, PCA/cPCA→t-SNE	beats t-SNE controls; cPCA better
MNIST-on-grass	real, additive	residual	the above, and raw	beats raw and t-SNE; ties cPCA
MNIST-on-grass, ten digits	real, additive	residual	the above	beats raw, t-SNE and label-free cPCA; label-tuned cPCA better

The main gaps, in rough order of importance:

1. **Non-additive real data.** Every real-data result is on an additive dataset, where global cPCA is exact in population. This is also where the local residual graph's case against a global cPCA graph must be made (section 5.5).
2. **Continuous or curved backgrounds.** Every background so far consists of discrete, well-separated clusters.
3. **The role of background negatives.** With the residual graph they add little on MNIST-on-grass, with two or ten digits, and are untested elsewhere, for example when the residual graph is imperfect.
4. **Label-free defaults beyond MNIST-on-grass.** k_{loc} and r have been checked on one dataset only.
5. **The algorithm confound.** Comparisons with t-SNE-based controls also change the embedding algorithm. Without the background term, Neg-t-SNE was below t-SNE on MNIST-on-grass (section 5.1).
6. **Nonlinear contrastive methods**, such as contrastiveVI [13], have not been compared.
7. **Scaling.** Only $n = 5000$ has been tested. The cost is in the same class as standard neighbor embeddings (section 4.4), but run times at larger n have not been measured.

8 Proposed next steps

1. **A real dataset with a non-additive background,** with BC-NE (with or without background negatives) as the main method, compared against cPCA→t-SNE, t-SNE and PCA→t-SNE. The dataset is to be chosen together. It should be one where the nuisance interacts with the signal, so that global cPCA's identity fails.
2. **What background negatives are for.** Test whether they help when the residual graph is imperfect, for example with a mis-sized k_{loc} or r. If they never help, drop them from the method.

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A Map to the repository

The repository (section B) organizes the work by phases. Each phase has its own detailed report and results directory, and runs carry short internal labels. The table below maps the names used here to those phases and files, for readers who want more detail; the report does not depend on them.

name in this report	internal label	detailed report	results directory
additive benchmarks (orthogonal / overlapping / nonlinear), uniform control, spectrum reference	easy / medium / hard; runs (b), (d)	`REPORT.md` (Phase 4)	`results/phase4/`
cPCA→Neg-t-SNE on the additive benchmarks	(e2)	`REPORT_phase5prep.md`	`results/phase5prep/`
MNIST-on-grass, raw graph; t-SNE / PCA→t-SNE / cPCA→t-SNE	(c), (a); T1 / T2 / T3	`REPORT_mnist_grass.md`	`results/mnist_grass/`
BC-NE / BC-NE+BN; residual-graph grid	(r_a) / (r_c); Phase 1	`REPORT_bsgraph.md`	`results/bsgraph/`
MNIST-on-grass with ten digits	(r_c), (r_a), (c), (a); T1–T3	`REPORT_mnist_grass_0to9.md`	`results/mnist_grass_0to9/`
two-stage vs one run, without background negatives	r_a-single	`REPORT_single_stage.md`	`results/single_stage/`
plain Neg-t-SNE	(a)	all	—
raw graph + background negatives	(c)	all	—
background-negative distribution	ξ_B	`REPORT.md`	`results/phase2/`

`NOTES.md` documents the package internals: the sampling scheme, the classifier constant and the schedule facts. The original plan is in `Initial_Project_Plan.md`.

B Reproducibility

Code. The code is at `https://github.com/Cassiebastress/bg-contrastive-ne` (public). Every result file records its software environment and the commit it ran at.

Pre-registration commits. Each was made before the results it governs.

experiment	commit
additive benchmarks: protocol and decision rules	`dad1c09`
global metrics and thresholds, before any cPCA result	`23459ff`
MNIST-on-grass (raw graph): pre-registration; selection	`8f916ee`; `3dd613a`
residual graph: graph-quality gate; embeddings	`3f0fc27`; `42bd31c`
MNIST-on-grass, ten digits: pre-registration; selection	`a9329a6`; `9a3ec00`
two-stage vs one run: tolerances	`d6f7be0`

Environment.

- Python 3.12, PyTorch 2.14.1 (CPU), numpy 2.5.2, scipy 1.18.1 and scikit-learn 1.9.1.
- `contrastive-ne` v0.4.0 (upstream commit `d5f0b53`), vendored with an 11-line hook.
- openTSNE 1.0.4 [11], for the t-SNE controls.

Compute.

- **Synthetic experiments:** an Apple M3 laptop, 4 threads.
- **MNIST-on-grass:** the McGill Mathematics Slurm cluster (AMD EPYC 9474F, CPU only), 1 thread per run, 24 runs in parallel.
- **Cross-platform check:** no comparison in this report mixes the two platforms. Before use, the cluster was validated: two runs were bit-identical. On a synthetic benchmark rerun, it differed from the laptop by at most 0.027 in kNN accuracy, below every verdict margin *(descriptive)*.
- **Determinism:** deterministic PyTorch algorithms, fixed thread counts, seeded generators and exact kNN graphs, with a bit-identical rerun checked in every experiment.

C Additional tables and figures

Table 7: Residual-graph grid on MNIST-on-grass, selection seeds 0–2, mean [min, max]. Selected: $k_{\text{loc}} = 30$, $r = 2$. The rule was the highest mean; variants within 0.002 counted as tied, and ties were broken toward the smaller r, then the smaller k_{loc}. Edge purity is the share of graph edges joining same-digit images.

graph	graph digit acc.	edge purity	grass recall	clean-digit recall
$k_{\text{loc}} = 15$, $r = 0$	0.960 [0.959, 0.961]	0.945 [0.943, 0.945]	0.084 [0.083, 0.084]	0.062 [0.060, 0.063]
$k_{\text{loc}} = 15$, $r = 1$	0.959 [0.958, 0.960]	0.948 [0.947, 0.950]	0.066 [0.065, 0.068]	0.064 [0.061, 0.066]
$k_{\text{loc}} = 15$, $r = 2$	0.961 [0.959, 0.962]	0.950 [0.949, 0.951]	0.057 [0.054, 0.059]	0.065 [0.063, 0.067]
$k_{\text{loc}} = 15$, $r = 5$	0.961 [0.958, 0.962]	0.951 [0.950, 0.954]	0.049 [0.047, 0.051]	0.068 [0.065, 0.070]
$k_{\text{loc}} = 15$, $r = 10$	0.958 [0.955, 0.961]	0.951 [0.949, 0.952]	0.046 [0.045, 0.049]	0.068 [0.066, 0.071]
$k_{\text{loc}} = 30$, $r = 0$	0.959 [0.956, 0.961]	0.941 [0.938, 0.942]	0.096 [0.093, 0.098]	0.060 [0.057, 0.062]
$k_{\text{loc}} = 30$, $r = 1$	0.960 [0.959, 0.962]	0.946 [0.944, 0.947]	0.073 [0.072, 0.073]	0.063 [0.059, 0.065]
$k_{\text{loc}} = 30$, $r = 2$	0.962 [0.961, 0.963]	0.950 [0.949, 0.951]	0.056 [0.054, 0.058]	0.066 [0.063, 0.068]
$k_{\text{loc}} = 30$, $r = 5$	0.961 [0.960, 0.962]	0.953 [0.952, 0.954]	0.046 [0.043, 0.049]	0.070 [0.067, 0.073]
$k_{\text{loc}} = 30$, $r = 10$	0.958 [0.955, 0.960]	0.951 [0.950, 0.954]	0.045 [0.041, 0.049]	0.071 [0.068, 0.074]
$k_{\text{loc}} = 100$, $r = 0$	0.954 [0.951, 0.955]	0.930 [0.926, 0.931]	0.125 [0.122, 0.130]	0.054 [0.050, 0.056]
$k_{\text{loc}} = 100$, $r = 1$	0.958 [0.956, 0.960]	0.937 [0.934, 0.939]	0.097 [0.094, 0.101]	0.057 [0.054, 0.060]
$k_{\text{loc}} = 100$, $r = 2$	0.963 [0.962, 0.964]	0.948 [0.947, 0.950]	0.061 [0.059, 0.064]	0.065 [0.062, 0.068]
$k_{\text{loc}} = 100$, $r = 5$	0.964 [0.962, 0.965]	0.957 [0.955, 0.958]	0.043 [0.040, 0.047]	0.074 [0.069, 0.076]
$k_{\text{loc}} = 100$, $r = 10$	0.959 [0.958, 0.959]	0.954 [0.953, 0.956]	0.044 [0.040, 0.047]	0.075 [0.071, 0.079]
raw pixels	0.904 [0.903, 0.905]	0.844 [0.837, 0.851]	0.439 [0.432, 0.447]	0.020 [0.019, 0.023]
PCA-50	0.935 [0.932, 0.938]	0.872 [0.867, 0.878]	0.337 [0.328, 0.348]	0.020 [0.019, 0.022]
global cPCA-10 (descriptive)	0.970 [0.947, 0.986]	0.956 [0.924, 0.978]	0.006 [0.006, 0.006]	0.065 [0.018, 0.103]

Table 8: Selection of m_B on MNIST-on-grass, selection seeds 0–2: digit kNN accuracy, mean [min, max]. The rule was the highest mean, with ties broken toward the smaller m_B; the selected values are in bold.

m_B	residual graph (BC-NE+BN)	raw graph + background negatives
0	0.972 [0.971, 0.974]	0.850 [0.818, 0.876]
1	**0.973** [0.971, 0.974]	0.923 [0.910, 0.931]
2	0.972 [0.971, 0.973]	0.935 [0.934, 0.936]
5	0.972 [0.969, 0.974]	**0.937** [0.936, 0.939]
10	0.970 [0.968, 0.972]	0.930 [0.928, 0.932]
20	0.969 [0.967, 0.971]	0.924 [0.919, 0.927]
40	0.968 [0.966, 0.969]	0.922 [0.919, 0.925]

Table 9: Additive benchmarks: pre-registered per-seed differences in signal kNN accuracy (background negatives minus control) on the confirmation seeds. All are consistent across 3 paired seeds *(pre-registered)*.

control	orthogonal	overlapping	nonlinear
plain Neg-t-SNE	$+0.589, +0.589, +0.568$	$+0.227, +0.197, +0.218$	$+0.504, +0.552, +0.507$
uniform, same m_B	$+0.543, +0.538, +0.541$	$+0.181, +0.169, +0.182$	$+0.472, +0.492, +0.464$
uniform, best m_B (20)	$+0.519, +0.489, +0.513$	$+0.181, +0.169, +0.182$	$+0.444, +0.439, +0.436$
spectrum reference	$+0.556, +0.547, +0.541$	$+0.206, +0.198, +0.189$	$+0.479, +0.508, +0.474$

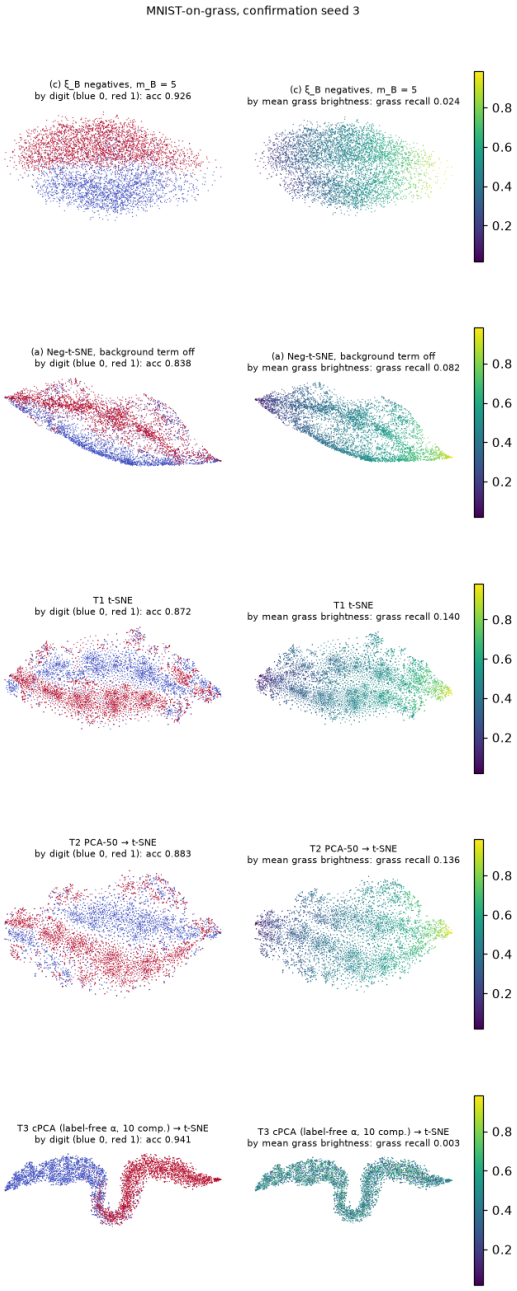


Figure 3: MNIST-on-grass controls, confirmation seed 3, colored by digit (left) and by grass brightness (right). Rows, top to bottom: raw graph + background negatives; plain Neg-t-SNE; t-SNE; PCA-50→t-SNE; cPCA→t-SNE. Plain Neg-t-SNE, t-SNE and PCA→t-SNE are organized partly by grass brightness, and their bright-grass end mixes the two digits.